Adding and Subtracting Rational Expressions

Name _____

Algebra 2 Lesson 4-4 (fast track) Date _____ Period _____

Learning goal: I can find the sum or difference of rational expressions and write a compound fraction in a simpler form.

Language goal: I can explain each step of adding, subtracting or simplifying using the words factor, least common denominator, numerator and compound fraction.

On the quiz: Questions 1 (full Savvas quiz only) and 3 add or subtract with different denominators: **Example 4** (factor, build the LCD, subtract the *whole* numerator). Question 2 (an LCM; full Savvas quiz only) is its first step. Questions 4 and 5 simplify a compound fraction: **Example 6**. The short form grades 3 and 4 (10 points each); 5 is a +5 bonus.

Example 4: Subtract Rational Expressions

What is the difference between $\dfrac{x+1}{x^2-6x-16}$ and $\dfrac{x+1}{x^2+6x+8}$?

$$= \frac{x+1}{(x-8)(x+2)} - \frac{x+1}{(x+2)(x+4)}$$

1. Factor each denominator.

$$= \frac{(x+1)(x+4)}{(x-8)(x+2)(x+4)} - \frac{(x-8)(x+1)}{(x-8)(x+2)(x+4)}$$

2. Build the LCD.

$$= \frac{(x^2+5x+4)-(x^2-7x-8)}{(x-8)(x+2)(x+4)}$$

3. Subtract the *whole* numerator.

$$= \frac{x^2+5x+4-x^2+7x+8}{(x-8)(x+2)(x+4)}$$

4. Distribute the -1.

$$= \frac{12x+12}{(x-8)(x+2)(x+4)} = \frac{12(x+1)}{(x-8)(x+2)(x+4)}$$

5. Combine and factor.

LCD: each different factor, to its highest power (here each appears once). No common factor, so the **answer** is $12(x+1)/\big((x-8)(x+2)(x+4)\big)$, $x \neq -4, -2, 8$.

Say it: First I _____ each denominator. The least common denominator is _____. When I subtract, I subtract the whole _____, so I distribute the _____.

Try It 4: Simplify

a. $\dfrac{1}{3x} + \dfrac{1}{6x} - \dfrac{1}{x^2}$

LCD = _____

Answer: _____ for $x \neq$ _____

b. $\dfrac{3x-5}{x^2-25} - \dfrac{2}{x+5}$

LCD = _____

Answer: _____ for $x \neq$ _____

Example 6: Simplify a Compound Fraction

A **compound fraction** has a fraction in its numerator, its denominator, or both (*fracción compuesta*). Write a simpler form of the compound fraction

$$\frac{\frac{1}{x}+\frac{2}{x+1}}{\frac{1}{y}}.$$

1. List every small denominator: x, $x+1$, y. Their LCM (the least common denominator) is $x(x+1)y$.

2. Multiply the whole numerator and the whole denominator by $\frac{x(x+1)y}{1}$. That is multiplying by 1, so the value does not change.

$$\frac{\left[\frac{1}{x}+\frac{2}{x+1}\right]\cdot x(x+1)y}{\frac{1}{y}\cdot x(x+1)y}=\frac{\frac{1}{x}\cdot x(x+1)y+\frac{2}{x+1}\cdot x(x+1)y}{\frac{1}{y}\cdot x(x+1)y}$$

3. Divide out each small denominator. Every small fraction is gone:

$$=\frac{(x+1)y+2xy}{x(x+1)}$$

4. Simplify the numerator: $(x+1)y+2xy=xy+y+2xy=3xy+y=(3x+1)y$.

$$=\frac{(3x+1)y}{x(x+1)}$$

Answer: $\frac{(3x+1)y}{x(x+1)}$ when $x\neq -1$, 0 and $y\neq 0$ (no small denominator can be 0).

Say it: This is a _____ because its numerator has fractions in it. I multiply the top and the bottom by the _____, which is _____. Then no small fractions are left, and I simplify the _____.

Try It 6: Simplify each compound fraction

a. $\dfrac{\frac{1}{x-1}}{\frac{x+1}{3}+\frac{4}{x-1}}$

Small denominators: _____

LCD = _____

_____

_____

_____

_____

_____

Answer: _____ for $x\neq$ _____

b. $\dfrac{2-\frac{1}{x}}{x+\frac{2}{x}}$

Small denominators: _____

LCD = _____

_____

_____

_____

_____

_____

Answer: _____ for $x\neq$ _____

Practice: Plug into a formula, then simplify

A car drives to a town at x miles per hour and drives back at $x + 10$ miles per hour. Its average speed for the round trip is

$$s = \frac{2}{\frac{1}{x} + \frac{1}{x+10}}.$$

a. The small denominators are _____. The LCD is _____. Simplify s.

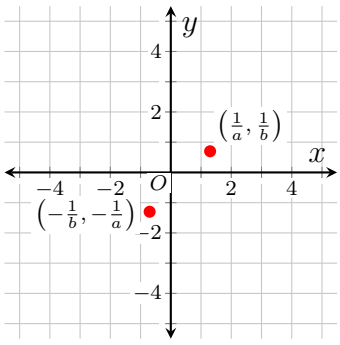
b. Use your answer: if $x = 40$, what is the average speed? Is it 45 mph?

IXL: Practice for the quiz

Code	Skill	Feeds
FEX ★	Add and subtract rational expressions	Quiz 3; Quiz 1 (full Savvas quiz only) (Example 4)
YD8 ★	Simplify complex fractions	Quiz 4 and 5 (Example 6)
UB5	Factor quadratics	Warm-up: step 1; Quiz 2 (LCM; full Savvas quiz only)
37N	Simplify rational expressions	Warm-up: the last step

SmartScore 70 or above on each starred (★) skill replaces your quiz score.

Exit question: Use structure (Practice & Problem Solving 14)



Find the slope of the line that passes through the points shown. Express it in simplest form.

$$\text{slope} = \frac{y_2 - y_1}{x_2 - x_1} = _____$$

Before the quiz, finish these sentences.

To add or subtract, I first _____ each denominator and find the _____.

When I subtract, I distribute the -1 to every term of the second _____.

To simplify a _____, I multiply its top and bottom by the LCD of all the small fractions.